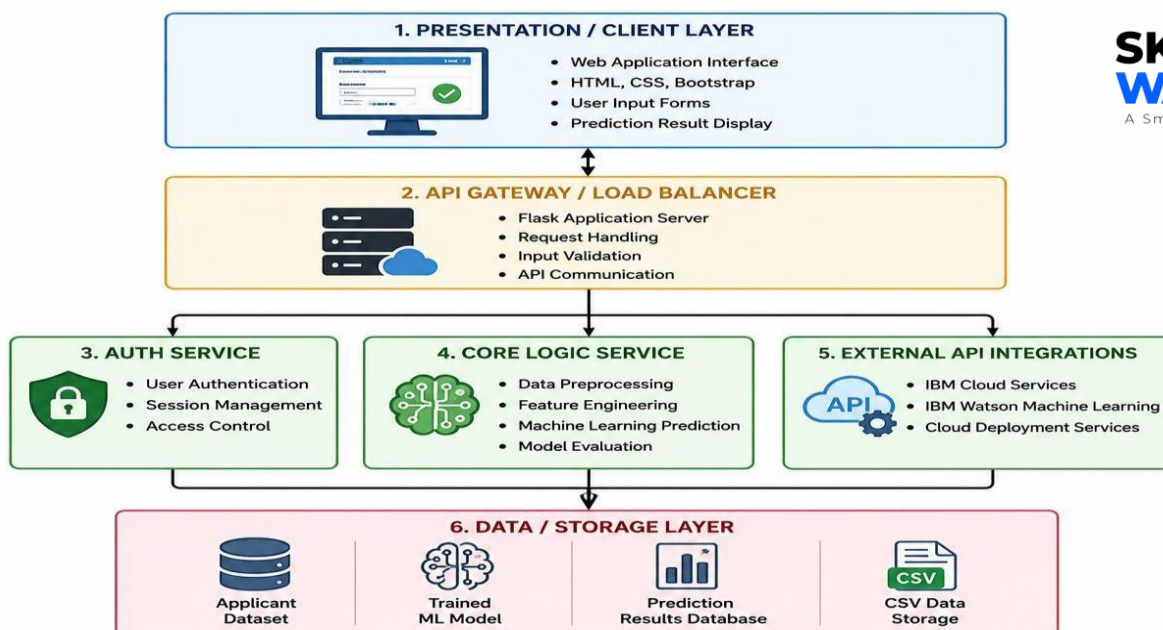


Solution Architecture

Date	7 July 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



COMPONENT DESCRIPTION TABLE

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects applicant details and displays prediction results	HTML, CSS, Bootstrap
API Gateway	Handles requests between frontend and backend	Flask
Microservices / Core Logic	Performs preprocessing and ML prediction	Python, Scikit-Learn, XGBoost
Database	Stores applicant data, model files, and prediction results	CSV Files, Pandas, IBM Cloud Storage



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where applicants enter their personal and financial details and view prediction results.	HTML5, CSS3, JavaScript, Flask (Jinja2 Templates)
API Gateway / Web Server	Handles HTTP requests, routes user actions, validates input, and communicates with the prediction engine.	Flask, Gunicorn
Core Logic Service	Processes applicant data, performs preprocessing, feature engineering, loads the trained model, generates predictions, and calculates confidence scores and recommendations.	Python, Pandas, NumPy, Scikit-learn, XGBoost, Joblib
Authentication Service	Not implemented because the application does not require user login or registration.	Not Applicable

External API Integrations	No third-party APIs are used. All predictions are generated locally using the trained machine learning model.	Not Applicable
Data / Storage Layer	Stores the dataset used for model training and the serialized machine learning artifacts required for prediction.	CSV, Joblib (.pkl Files)